# Plan

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## Five lectures on the MatterMost paper:

- The theory behind the MatterMost paper (four lectures)
- The main idea of the MatterMost paper (one lecture)

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## Today:

- The main idea of the MatterMost paper

# Responsibility under uncertainty: which climate decisions matter most?

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- Which decisions should not matter (for ...)?

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- Matter for what?
- What does it mean for a decision to matter (for . . . )?
- Which decisions should not matter (for . . . )?
- How does a measure of which decisions matter most relate to the notion of responsibility?

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- For example, one who encounters a car accident may be regarded as worthy of praise for having saved a child from inside the burning car, or alternatively, one may be regarded as worthy of blame for not having used one's mobile phone to call for help.
- To regard such agents as worthy of one of these reactions is to regard them as responsible for what they have done or left undone.

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- can come in different measures: more or less praise or blame.

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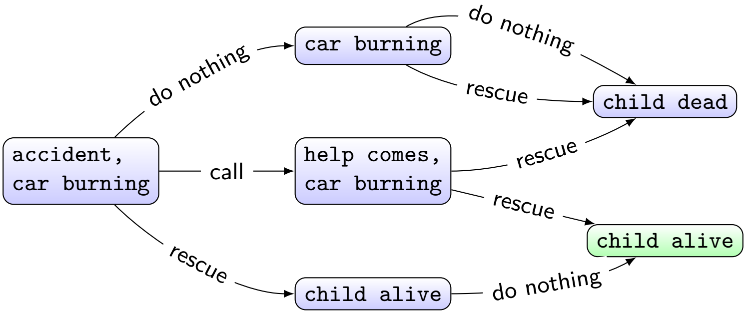

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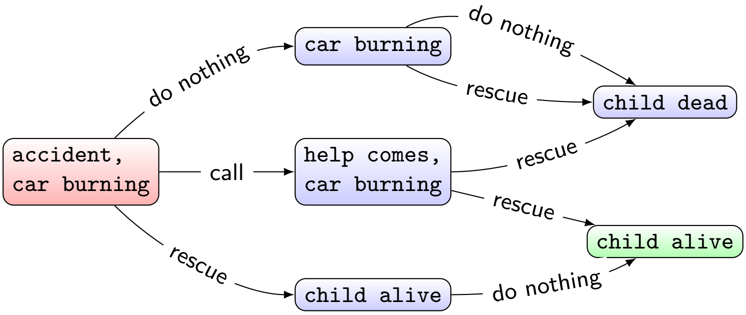
- In the first node, decisions seem to matter a lot: high responsibility?



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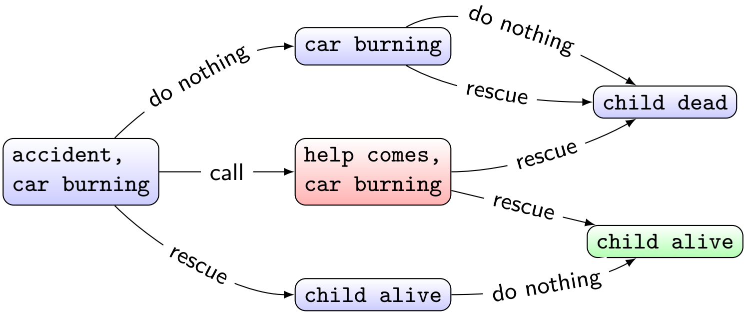
- What about the next node in the middle: low / high responsibility?



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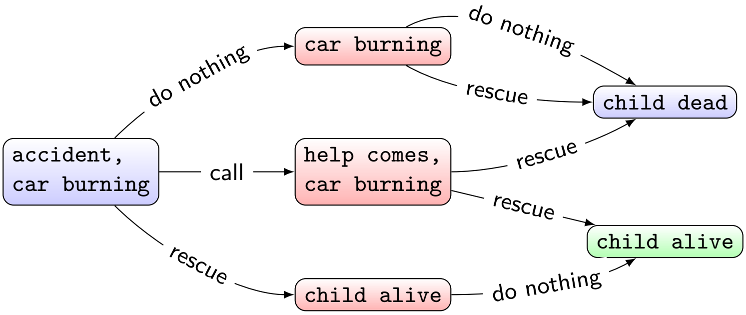
- All the same responsibility?



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- Agency: the capability to act intentionally, to plan, and to distinguish between desirable and undesirable outcomes.
- Avoidance: one shall not be held responsible for something that cannot be avoided.
- Causal relevance: there should be a causal relation between the conduct of the agent and the resultant outcome.

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- ...ex-ante (type, general, ...) and ex-post (actual, specific, ...) responsibility, see page 1 of *J. Halpern. Actual causality*.
- ...individual and collective responsibility.
- Our contribution is towards understanding and measuring ex-ante responsibility, no attempt at distinguishing between individual and collective responsibility.

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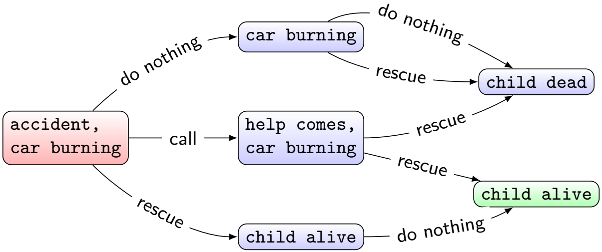

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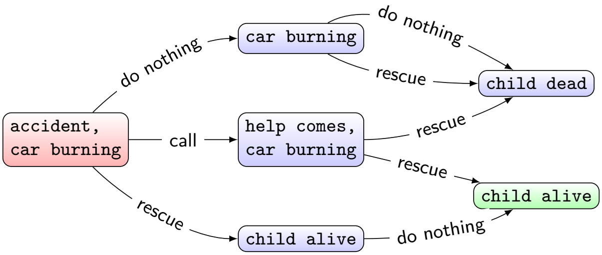


- The difference between the value of the best decision and that of the worst action is large!

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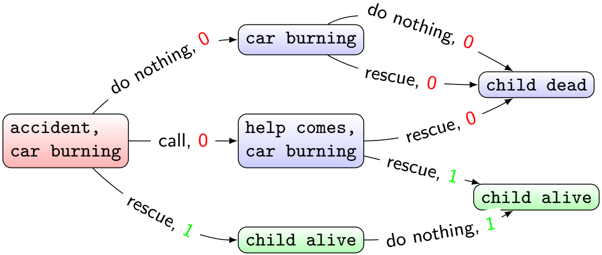

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## Exercise 8.1

Define *Singleton* in Agda. Under which other conditions do you expect *mMeas* to be zero? *mMeas* depends on $t$ and on $x$ but also on a number of decision steps $n$. Explain why.

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but we have to make assumptions about future decisions. Explain why. Define *worstVal* under the assumption that all future decisions will be taken optimally, that is $worstVal \; t \; (n + 1) \; x$ is the value of a worst decision at decision step $t$ in state $x$ given that all subsequent $n$ decisions are taken optimally.

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- These boil down to the observation that, in general, a "moral" approach towards decision making – doing the right thing even though the probability of success becomes increasingly small – is rational (it pays off) over a wide range of uncertainties.